Generative Models for Cat Face Synthesis: A Comparative Study of Variational Autoencoders, Generative Adversarial Networks, and Diffusion Models

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Abstract

This report presents a comparative study of three major generative modeling paradigms—Variational Autoencoders (VAE), Generative Adversarial Networks (GAN), and Diffusion Models—applied to the synthesis of 128×128 cat face images. We evaluate β-VAE and VQ-VAE-1 (VAE family), WGAN-GP and SN-GAN (GAN family), DDIM and a Tiny Latent Diffusion Model (diffusion family), and add a classical PixelCNN autoregressive prior over VQ-VAE code indices as a baseline for latent generative modeling. The study employs structured grid search over key hyperparameters. We evaluate models using Fréchet Inception Distance (FID), latent interpolations, and a qualitative comparison of PixelCNN versus Tiny LDM (sample quality and wall-clock generation time). Furthermore, we conduct an exploratory extension training of SN-GAN on mixed dogs-and-cats data to observe distribution averaging phenomena. This work provides a practical framework for comparing generative paradigms.

Index Terms: generative models, variational autoencoders, generative adversarial networks, diffusion models, PixelCNN, latent diffusion, image synthesis, cat generation, FID, hyperparameter optimization, mode collapse

1. Introduction

Generative modeling of images has been a central research challenge in machine learning, with three primary paradigms emerging over the past decade: Variational Autoencoders (VAEs) [1], Generative Adversarial Networks (GANs) [2], and Diffusion Models [3]. Each paradigm offers distinct theoretical foundations, training dynamics, and trade-offs in quality, diversity, and training stability. VAEs provide a principled probabilistic framework but often produce blurrier samples; GANs generate sharp images but suffer from training instability and mode collapse; diffusion models have recently demonstrated superior sample quality but require longer inference times.

The selection of appropriate generative models for practical applications remains highly dependent on specific constraints: available computational resources, target sample quality, required inference speed, and stability during training. In this project, we conduct a systematic comparison of representative architectures across all three paradigms on a well-defined task: generating cat face images from the Kaggle Cat Dataset.

Our objectives align with the specific project requirements: (i) standardize varying image resolutions utilizing raw annotation data to accommodate limited computing resources; (ii) systematically compare generation quality across three paradigms (VAE, GAN, Diffusion) using quantitative metrics (FID) and qualitative assessment; (iii) directly address mode collapse through stabilized GAN architectures (WGAN-GP with gradient penalty, SN-GAN with spectral normalization); (iv) analyze latent space smoothness via interpolation: generating 10-image sequences (2 random endpoints + 8 linear intermediates) to assess semantic continuity; and (v) explore cross-species generalization by training on mixed "Dogs vs. Cats" data to investigate whether the unconditional model learns to separate classes or exhibits distribution averaging (e.g., chimera-like outputs).

2. Methodology

2.1. Datasets and Preprocessing

The primary dataset consists of over 9,000 cat images sourced from the Kaggle Cat Dataset [4]. Raw images exhibit significant variability in resolution, aspect ratio, and face positioning, which poses a challenge for batch processing and limited computing resources. To address this, we apply the following preprocessing pipeline:

1. **Annotation Parsing**: Instead of relying on pre-trained face detectors, we extract the exact 9 coordinate points (eyes, mouth, ears) provided in the corresponding `.cat` annotation files.
2. **Face Cropping**: Bounding boxes are dynamically calculated based on the extreme coordinates of the facial landmarks with an added spatial margin to capture the full face.
3. **Resizing**: Cropped regions are standardized to 128×128 pixels using bilinear interpolation, drastically reducing VRAM footprint and training time.
4. **Normalization**: All cat-face training tensors are mapped to $[-1, 1]$ via `ToTensor` and per-channel normalization with mean 0.5 and std 0.5, matching `tanh` output heads on VAE decoders, GAN generators, and diffusion pipelines.

Comparison of raw vs. processed images is shown in Figure 1.

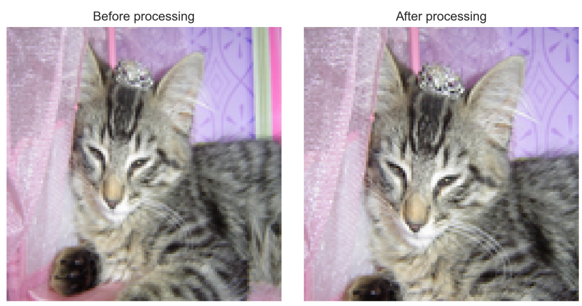

Figure 1: *Sample images after preprocessing: cropped and resized to* 128×128 *pixels.*

For the exploratory extension, we utilize the Kaggle "Dogs vs. Cats" competition benchmark dataset [5], which contains $\sim 25,000$ mixed images (Figure 2). Since this dataset lacks facial landmark annotations, applying the aforementioned coordinate-based cropping procedure is impossible. Instead, to prevent aspect ratio distortion that could negatively impact the FID metric, we apply a scale-and-crop strategy: resizing the shorter edge of the image to 128 pixels, followed by extracting a 128×128 center crop. This forms a secondary dataset designed to test cross-species distribution averaging.

Figure 2: *Sample images from the mixed "Dogs vs. Cats" dataset after scale-and-crop preprocessing.*

2.2. Generative Model Architectures

2.2.1. Variational Autoencoders (VAE)

VAEs define a probabilistic generative model via a learned latent variable distribution $q_\phi(z|x)$ parameterized by an encoder network. The generative process is specified by a decoder $p_\theta(x|z)$ where $z \sim \mathcal{N}(0, I)$. We evaluate two VAE variants:

β-VAE [6]: A standard VAE with convolutional encoder/decoder and a β parameter that controls the trade-off between reconstruction accuracy and prior matching. Grid search varies: latent_dim $\in \{64, 128\}$, $\beta \in \{1.0, 4.0\}$.

VQ-VAE-1: A discrete VAE using vector quantization in the latent space [7]. Instead of a continuous distribution, VQ-VAE discretizes the latent space into a learned codebook to reduce the blurriness typical of standard VAEs. Grid search varies: codebook_size $\in \{512, 1024\}$, feat_map_size $\in \{16 \times 16, 8 \times 8\}$, and reconstruction loss (L1 vs. L2).

2.2.2. Generative Adversarial Networks (GAN)

GANs formulate image generation as an adversarial game between a generator G and discriminator D. To directly address the mode collapse problem required by this study, we avoid the standard DCGAN and evaluate two stabilized variants:

WGAN-GP: Wasserstein GAN [8] with gradient penalty [9] uses the Wasserstein distance and adds a gradient penalty term to enforce the Lipschitz constraint, significantly reducing mode collapse and providing meaningful learning curves. Grid search varies: n_critic $\in \{3, 5\}$, batch size $\in \{64, 128\}$, learning rate schedule $\in \{\text{symmetric}, \text{TTUR}\}$.

SN-GAN: Spectral Normalization GAN [10] stabilizes discriminator training by normalizing weights with their largest singular value. Grid search varies: batch size $\in \{64, 128\}$, learning rate schedule $\in \{\text{symmetric}, \text{TTUR}\}$, horizontal flip augmentation.

2.2.3. Diffusion Models

Diffusion models define a forward process that corrupts images to noise, then learn to reverse this process via a neural network. To mitigate the immense computational cost of diffusion, we focus on optimized architectures:

DDIM: A faster sampling approach for diffusion models [11] that uses a deterministic sampling procedure, enabling inference in fewer steps. Grid search varies: noise schedule $\in \{\text{linear}, \text{cosine}\}$, base channels $\in \{32, 64\}$.

Tiny Latent Diffusion Model (LDM): Following Rombach et al. [12], we train a lightweight U-Net diffusion model in the *frozen* latent space of the best VQ-VAE-1 from the grid search. The VQ encoder/decoder weights are not updated during LDM training; only the U-Net is optimized. At inference, DDIM sampling produces a latent tensor that is passed through the frozen decoder. Grid search varies DDIM steps $\in \{50, 100, 200\}$ at inference (no retraining per step count).

2.2.4. PixelCNN Prior over VQ Codes (Baseline)

VQ-VAE-1 alone defines $p(x|z)$ via the decoder but not a prior $p(z)$ over discrete code indices. For unconditional synthesis we therefore compare two learned priors on the same frozen VQ-VAE:

PixelCNN [13]: A lightweight autoregressive model with masked convolutions predicts codebook indices on the 16×16 or 8×8 latent grid (depending on the VQ-VAE configuration).

2.3. Training Setup and Hyperparameter Grid

All models employ the Adam optimizer with lr $= 2 \times 10^{-4}$ and $(\beta_1, \beta_2) = (0.5, 0.999)$. Training uses a high epoch cap (1000) with early stopping (patience $= 15$, min_epochs $= 20$, min_delta $= 10^{-4}$). β-VAE and VQ-VAE early stopping monitor **reconstruction loss** rather than total loss (which includes KL or VQ terms that can plateau early).

Each hyperparameter configuration is trained independently across three fixed random seeds $\{42, 0, 3407\}$. Seeds are reset in PyTorch, NumPy, and Python before every run. Grid-sweep results are compared per cell; evaluation metrics are aggregated over the same seed list for reproducibility.

Table 1: *Hyperparameter grid summary across generative model families.*

Model Family	Parameter	Search Space
β-VAE	latent_dim, β	$\{64, 128\} \times \{1.0, 4.0\}$
VQ-VAE-1	codebook, map, loss	$\{512, 1024\} \times \{16, 8\} \times \{L1, L2\}$
WGAN-GP	n_critic, batch, LR	$\{3, 5\} \times \{64, 128\} \times \{sym, TTUR\}$
SN-GAN	batch, LR, augment	$\{64, 128\} \times \{sym, TTUR\} \times \{no, hflip\}$
DDIM	schedule, channels	$\{linear, cosine\} \times \{32, 64\}$
Tiny LDM	steps (inference)	$\{50, 100, 200\}$
PixelCNN prior	hidden width, depth	128 channels, 10 masked layers

2.4. Evaluation Metrics

2.4.1. Fréchet Inception Distance (FID)

FID compares the distribution of real validation images to unconditionally generated samples in the feature space of a pretrained Inception network. Lower FID indicates better alignment with the data distribution. We use each model's native sampling procedure (e.g., $z \sim \mathcal{N}(0, I)$ for β-VAE and GANs, DDIM for pixel diffusion, random codes for VQ-VAE when no separate prior is evaluated). Generated tensors are mapped from $[-1, 1]$ to $[0, 1]$ before feature extraction, consistent with training normalization.

We report FID on the held-out validation set for every trained family: β-VAE, VQ-VAE-1, PixelCNN, Tiny LDM, WGAN-GP, SN-GAN, and DDIM, using a fixed sample count ($N{=}1000$) per run. After selecting a hyperparameter configuration from the sweep (shared VQ settings for VQ-VAE, PixelCNN, and Tiny LDM), we compute FID independently for each seed and summarize mean $\pm$ standard deviation. PixelCNN and Tiny LDM use their learned priors with the same frozen decoder; VQ-VAE FID uses the trainer's random-code debug sampler and is reported only as a weak baseline, not comparable to the learned priors.

2.4.2. Latent Interpolation Analysis

For models with a continuous latent vector $z \in \mathbb{R}^d$, we test whether the learned representation is semantically smooth. We sample two endpoints $z_1, z_2 \sim \mathcal{N}(0, I)$ from the prior used at generation time, form eight uniformly spaced convex combinations $z_\alpha = (1 - \alpha)z_1 + \alpha z_2$, $\alpha \in \{0, \frac{1}{9}, \ldots, 1\}$, and decode each z_α to an image, yielding a strip of ten frames. Continuity of appearance along the strip is judged qualitatively. This protocol is applied to β-VAE and WGAN-GP, which admit Gaussian latent walks; it is not used for discrete VQ codes or diffusion latents without an explicit interpolation definition.

2.4.3. Exploratory Extension: Mixed Distributions

We train Spectral Normalization GAN (SN-GAN) on a mixed dataset containing both cats and dogs, utilizing a preprocessed dataset of $\sim 25,000$ combined images. The goal is to observe whether the unconditional model learns to separate the classes into mode-specific outputs or suffers from distribution averaging, resulting in generating "chimeras" (e.g., cat faces with dog ears or dog faces with feline features). This experiment probes the limits of unconditional generation under multi-modal input distributions. SN-GAN was deliberately chosen for this task. Diffusion models were too computationally expensive for the $\sim 25,000$ image dataset, and VAEs were rejected because their tendency to blur reconstructions would obscure the distribution averaging effect. SN-GAN provides the sharp, detailed outputs necessary to clearly observe cross-class features in the generated samples.

3. Results

[To be populated]

4. Conclusions

[To be populated]

5. AI Disclosure

This report was edited with AI assistance, to improve clarity, consistency, and formatting. Part of the docstrings, some phrasing and code refactoring or debugging were refined using AI tools, but all scientific decisions and interpretations are the authors' own. The core content, experimental design, and analysis were developed by the authors.

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A. Appendix

A.1. Additional Experimental Details

[To be populated]

A.2. Code Availability

All code, configurations, and training logs are available at: `https://github.com/FlorekPawel/gen-cats`